CHAPTER 01

SOME BASIC CONCEPTS OF CHEMISTRY

CATHODE RAYS - DISCOVERY OF ELECTRON

- Electrons were opdiscovered using a gas-discharge tube experiment, where invisible rays, called cathode rays, travelled from the negative electrode (cathode) to the positive one (anode).
- These rays are streams of negatively charged particles (electrons) because they are deflected by electric and magnetic fields.
- The charge-to-mass ratio (e/m) for these particles is the same, no matter what gas is used in the tube.

PROPERTIES OF ELECTRON

- The charge of an electron was precisely measured by Robert Millikan's oil drop experiment.
- An electron has a charge of -1.6022×10^{-19} coulombs and a mass of 9.1096×10^{-31} kg.

POSITIVE RAYS-DISCOVERY OF PROTON

- Protons were discovered in a similar experiment using a perforated cathode, which allowed positive rays (canal rays) to pass through.
- These rays consist of positively charged particles.
- Unlike cathode rays, their charge-to-mass (e/m) ratio changes depending on the gas used in the tube.

PROPERTIES OF PROTON

- A proton has a positive charge of $+1.602 \times 10^{-19}$ C and a mass of 1.672×10^{-27} kg, which is about one atomic mass unit (amu).

DISCOVERY OF NEUTRON

- In 1932, Chadwick discovered the neutron by bombarding a sheet of beryllium with alpha particles.
- This process produced neutral particles with a mass similar to that of a proton.
- The nuclear.reaction was: $^{4}He + {}^{9}Be \rightarrow {}^{12}C + {}^{1}n$.

PROPERTIES OF NEUTRON

- A neutron is a subatomic particle with no electrical charge and a mass of approximately 1 amu.
- Because it has no charge, its specific charge (e/m value) is zero.

THOMSON'S MODEL OF ATOM

- Thomson proposed the "plum pudding" model, suggesting that an atom is a uniform sphere of positive charge with negatively charged electrons embedded within it.
- This model was later proven incorrect as it could not explain the results of Rutherford's scattering experiment.

RUTHERFORD'S SCATTERING EXPERIMENT

- Rutherford fired positively charged alpha (α) particles at a thin gold foil.
- Most particles passed straight through, a few were deflected, and a very small number bounced back.
- This showed that an atom is mostly empty space with a tiny, dense, positively charged centre called the nucleus, where almost all the mass is located.

DISTANCE OF CLOSEST APPROACH

- Rutherford estimated the size of the nucleus by calculating how close an alpha particle could get before being repelled by the positive nuclear charge.
- At this point, the particle's initial kinetic energy (K.E.) is completely converted into electrostatic potential energy.
- Formula: $r = (2Ze^2) / (4\pi\varepsilon_0 \times K.E.)$

APPLICATIONS OF RUTHERFORD MODEL

- Based on his experiment, Rutherford proposed the nuclear model of the atom, where electrons orbit a central nucleus.
- This model established that the radius of an atom is about 100,000 times larger than the radius of its nucleus.

DRAWBACKS OF RUTHERFORD MODEL

- According to classical physics, an orbiting electron should continuously lose energy as radiation, causing it to spiral into the nucleus.
- This meant the model could not explain why atoms are stable.

ATOMIC NUMBER (Z) AND MASS NUMBER (A)

- **Atomic Number (Z):** The number of protons in an atom's nucleus. It defines the element.
- **Mass Number (A):** The total number of protons and neutrons in the nucleus.
- **Isotopes** have the same number of protons but different numbers of neutrons (same Z, different A). **Isobars** have the same mass number but different numbers of protons (same A, different Z).

NATURE OF LIGHT

- James Maxwell's theory describes light as an electromagnetic wave made of oscillating electric and magnetic fields.

These fields are perpendicular to each other and to the direction the wave is travelling.

Electromagnetic waves can travel through a vacuum at the speed of light ($c = 3 \times 10^8$ m/s).

ME IMPORTANT CHARACTERISTICS OF A WAVE:

Wavelength (λ): The distance between two consecutive peaks (crests) or valleys (troughs) of a wave.

Frequency (ν): The number of waves that pass a point in one second.

The relationship between speed, wavelength, and frequency is: $c = \lambda\nu$.

LECTRO MAGNETIC SPECTRUM

This is the arrangement of all types of electromagnetic radiation in order of increasing wavelength or decreasing frequency.

The order includes gamma rays, X-rays, ultraviolet, visible light, infrared, microwaves, and radio waves.

PLANCK'S QUANTUM THEORY

- Planck suggested that energy is not emitted or absorbed continuously, but in discrete packets called "quanta" (or "photons" for light).
- The energy of one quantum is given by the formula $E = h\nu$, where h is Planck's constant.
- The total energy transferred is always a whole-number multiple of a quantum ($E = nh\nu$).

BLACK BODY RADIATION:

- A black body is an ideal object that absorbs and emits all frequencies of radiation. The spectrum of light it emits depends only on its temperature.
- Classical wave theory could not explain this phenomenon, but Planck's quantum theory successfully did.

PHOTOELECTRIC EFFECT:

- This is the emission of electrons from a metal surface when light shines on it.
- Einstein used Planck's theory to explain that light consists of photons, and one photon can eject one electron if it has enough energy.

TERMS USED IN PHOTO ELECTRIC EFFECT

- **Work function (w):** The minimum energy required to remove an electron from the surface of a particular metal.
- **Threshold frequency (ν_0):** The minimum frequency of light needed to eject an electron. If the light's frequency is below this, no electrons are emitted.

SATURATION CURRENT & STOPPING POTENTIAL

- **Stopping potential (V_0):** The minimum voltage required to stop the most energetic photoelectrons from reaching the other electrode.
- The maximum kinetic energy of the photoelectrons is directly related to the stopping potential: $K.E._{max} = eV_0$.

EFFECT OF VARIATION OF FREQUENCY

- The maximum kinetic energy of the ejected electrons increases linearly as the frequency of the incident light increases.
- Einstein's photoelectric equation summarises this: $K.E._{max} = h\nu - w$.

EFFECT OF VARIATION OF PHOTO INTENSITY

- Increasing the intensity (brightness) of the light increases the number of photons hitting the metal per second.
- This results in more electrons being ejected per second (a higher photocurrent), but it does not change the maximum kinetic energy of each electron.

POSTULATES OF BOHR'S MODEL

- Electrons move in fixed, circular orbits around the nucleus, called stationary states, without losing energy.
- The angular momentum of an electron is quantised, meaning it can only have specific values: $mvr = n(h/2\pi)$, where n is an integer (1, 2, 3...).
- An electron emits or absorbs energy as a photon only when it jumps from one orbit to another.

DERIVATION OF RADIUS OF SHELL

- The radius of an electron's orbit in a single-electron atom is given by the formula: $r = 0.529 \times (n^2/Z)$ Å.

DERIVATION OF VELOCITY OF ELECTRON IN BOHR'S ORBIT

- The velocity of an electron in a given orbit is calculated as: $v = 2.18 \times 10^6 \times (Z/n)$ m/s.

DERIVATION OF TOTAL ENERGY OF ELECTRON/SYSTEM

- The total energy of an electron in an orbit is the sum of its kinetic and potential energies.
- The energy is quantised and given by the formula: $T.E. = -13.6 \times (Z^2/n^2)$ eV/atom.
- The negative sign indicates that the electron is bound to the nucleus.

DEFINITIONS VALID FOR SINGLE ELECTRON SYSTEM

- **Ground State:** The lowest possible energy level for an electron (n=1).
- **Excited State:** Any energy level higher than the ground state (n > 1).
- **Ionisation Energy (I.E.):** The minimum energy needed to completely remove an electron from an atom in its ground state.

FAILURES/LIMITATIONS OF BOHR'S THEORY

- Bohr's model works perfectly for single-electron systems (like H or He^+) but fails to predict the properties of atoms with more than one electron.
- It could not explain the splitting of spectral lines in magnetic (Zeeman effect) or electric (Stark effect) fields.

SPECTRUM

- An **emission spectrum** consists of bright lines of specific colours, produced when excited atoms release energy as light.
- An **absorption spectrum** consists of dark lines in a continuous spectrum, formed when atoms absorb specific wavelengths of light.
- These unique line spectra act as "fingerprints" for elements.

LINE SPECTRUM OF H-ATOM

- The emission spectrum of hydrogen is not continuous but consists of distinct lines grouped into series.
- The wavelength (λ) of these lines can be calculated with the Rydberg formula: $1/\lambda = R_H (1/n_1^2 - 1/n_2^2)$.

HYDROGEN SPECTRAL SERIES

- The different series in the hydrogen spectrum are named after their discoverers and correspond to electrons falling to a specific final energy level (n_1).
- **Lyman Series:** Transitions to $n_1=1$ (in the Ultraviolet region).
- **Balmer Series:** Transitions to $n_1=2$ (in the Visible region).
- **Paschen, Brackett, and Pfund Series:** Transitions to $n_1=3$, 4, and 5, respectively (all in the Infrared region).

DUAL NATURE OF MATTER (DE-BROGLIE HYPOTHESIS)

- De Broglie proposed that all moving particles, including electrons, have wave-like properties (wave-particle duality).
- The de Broglie wavelength of a particle is given by $\lambda = h/p$, where h is Planck's constant and p is the particle's pmomentum (mv).

HEISENBERG'S UNCERTAINTY PRINCIPLE

- This principle states that it is impossible to know both the exact position and the exact momentum of a particle at the same time.
- The more precisely one is known, the less precisely the other can be known ($\Delta x \cdot \Delta p \geq h/4\pi$).
- This fundamental limit means we cannot think of electrons in fixed orbits, but rather in terms of probability clouds called orbitals.

THE QUANTUM/WAVE MECHANICAL MODEL

- This model uses the Schrödinger equation to describe the behavior of electrons in atoms.
- Solving the equation gives a **wave function (ψ)**, which contains all the information about the electron.
- The square of the wave function (ψ^2) represents the probability density of finding the electron at a certain location. An **orbital** is a region of space with a high probability of finding the electron.

RADIAL WAVE FUNCTIONS AND PROBABILITY

- A **radial node** is a spherical surface around the nucleus where the probability of finding an electron is zero. The number of radial nodes is given by (n - l - 1).
- The radial probability distribution plot shows the most probable distance from the nucleus to find an electron for a given orbital.

QUANTUM NUMBER

- Four quantum numbers are used to describe the location and properties of an electron in an atom.
- **Principal Quantum Number (n):** Defines the electron's main energy level or shell (n = 1, 2, 3...).
- **Azimuthal Quantum Number (l):** Defines the shape of the orbital and subshell (l = 0 for s, 1 for p, 2 for d...). Its value ranges from 0 to n-1.
- **Magnetic Quantum Number (m_l):** Describes the orientation of the orbital in space (e.g., p_x, p_y, p_z). Its value ranges from -l to +l.
- **Spin Quantum Number (m_s):** Describes the electron's intrinsic spin, which can be either "up" (+1/2) or "down" (-1/2).

RULES FOR FILLING SUBSHELL

- **Aufbau Principle:** Electrons fill the lowest energy orbitals first. The order can be remembered using a diagram.

(n+l) Rule: Orbitals with a lower (n+l) value are filled first. If the value is the same, the orbital with the lower n is filled first.

Pauli Exclusion Principle: No two electrons in an atom can have the same four quantum numbers. This means an orbital can hold a maximum of two electrons, which must have opposite spins.

Hund's Rule: When filling orbitals of equal energy, electrons occupy separate orbitals with parallel spins before pairing up.

EXTRA STABILITY OF HALF-FILLED & FULLY FILLED SUBSHELLS

- Subshells that are exactly half-filled (like d^5) or completely filled (like d^{10}) are exceptionally stable.
- This extra stability is due to symmetrical electron distribution and a phenomenon called exchange energy.
- This explains why the electron configurations of elements like chromium (Cr) and copper (Cu) are exceptions to the standard Aufbau principle.

PRACTICE QUESTIONS

ıgle Correct Type Questions

Container A = 100 g of 109% oleum
Container B = 100 g of 118% oleum
Ratio of % of free SO_3 in container A and B
(1) 0.5 (2) 1
(3) 1.5 (4) 3

Calculate the weight of FeO produced from 6.7 g VO & 4.8 g Fe_2O_3
$2VO + Fe_2O_3 \longrightarrow 2FeO + V_2O_5$
(At. wt. of V = 51, At. wt. of Fe = 56)
(1) 4.32 (2) 7.755
(3) 2.585 (4) 0.0718

3. The correct expression relating molality (m), molarity (M), density of solution (d) and molar mass (M_2) of solute is:

(1) $m = \dfrac{M}{d + MM_2} \times 1000$

(2) $m = \dfrac{M}{1000d - MM_2} \times 1000$

(3) $m = \dfrac{d + MM_2}{M} \times 1000$

(4) $m = \dfrac{1000d - MM_2}{M} \times 1000$

4. In Haber's process, N_2 combines with H_2 according to the reaction,
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$;
In which of the following reaction mixtures, will N_2 behave as the limiting reagent?
(1) 0.5 g-molecule of N_2 + 2 g- molecule of H_2
(2) 28 g of N_2 + 6 g of H_2
(3) 56 g of N_2 + 10 g of H_2
(4) 84 mg of N_2 + 18 mg of H_2

5. The density of a liquid is 1.2 g/mL. There are 35 drops in 2 mL. The number of molecules in 1 drop is [Molecular weight of liquid = 70 u and N_A = Avogadro number]

(1) $\dfrac{12}{35}N_A$ (2) $\left[\dfrac{1}{35}\right]^2 N_A$

(3) $\dfrac{12}{(35)^2}N_A$ (4) $\dfrac{1.2}{(35)^2}N_A$

6. 6.02×10^{20} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is-
(1) 0.001 M (2) 0.01 M
(3) 0.02 M (4) 0.1 M

7. Starting with 2 moles of A and 1 mole of B, the following reaction
$2A + 3B \longrightarrow X + 2Y$
is made to take place. Assume the reaction to go to completion. Match the number of moles listed in Column-II with various species listed in Column-I.

Column - I		Column - II	
A	A	P	1/3
B	B	Q	2/3
C	X	R	0.0
D	Y	S	4/3

	A	B	C	D
(1)	S	R	P	Q
(2)	P	Q	R	S
(3)	Q	S	R	P
(4)	Q	R	S	P

8. By complete combustion of 3.60 g of an oxygen containing compound ($C_xH_yO_z$), 5.28 g of CO_2 and 2.16g of H_2O were obtained. The empirical formula of the compound could be.
(1) C_2H_2O (2) C_2H_4O
(3) CH_2O (4) CH_4O_2

9. The percent composition of a highly explosive compound is

C	H	N	O
x%	1.34%	y%	42.65%

The molar mass of compound is 450.22 and x = 2y, then molecular formula is

(1) $C_9H_3N_5O_{11}$ (2) $C_{14}H_6N_6O_{12}$

(3) $C_{14}H_6N_7O_{12}$ (4) $C_{14}H_3N_6O_{12}$

10. The molar ratio of Fe^{2+} to Fe^{3+} in a mixture of $FeSO_4$ and $Fe_2(SO_4)_3$ having equal number of sulphate ion in both ferrous and ferric sulphate is

(1) 1 : 2 (2) 3 : 2

(3) 2 : 3 (4) Can't be determined

11. If given sample of potassium chlorate ($KClO_3$) is 80% pure, then 48 g of oxygen would be produced from

(1) 98 g of the sample

(2) 153.12 g of the sample

(3) 245 g of the sample

(4) 122.5 g of the sample

12. Consider a reaction of antimony with sulphur

$$2Sb(s) + 3S(s) \rightarrow Sb_2S_3(s)$$

If 1g of Sb_2S_3(s) is obtained from the reaction of 2 g of antimony and excess of sulphur, the % yield of the reaction is :

(Atomic mass of Sb = 122 u)

(1) 50% (2) 36%

(3) 30% (4) 26%

13. Zinc blende (ZnS) is roasted and then reduced to produce metallic Zn via the following series of reactions

$$ZnS + \frac{3}{2}O_2 \rightarrow ZnO + SO_2 \text{ (80\% yield)}$$

$$ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O \text{ (100\%)}$$

$$ZnSO_4 + Mg \rightarrow Zn + MgSO_4 \text{ (80\%)}$$

How many moles of ZnS are required to produce 3.2 moles of Zn?

(1) 4.5 (2) 5.0

(3) 6.25 (4) 8.0

14. Mole fraction of a solute in an aqueous solution is 0.2. The molality of the solution will be;

(1) 13.88 m (2) 1.388 m

(3) 0.138 m (4) 0.0138 m

15. The weight of one atom of calcium (40 amu) is approximately:

(1) 6.64×10^{-23} g (2) 4.0×10^{-24} g

(3) 1.66×10^{-22} g (4) 3.32×10^{-25} g

16. A compound XY contains 16 g of X and 24 g of Y . Another sample of the compound contains 40 g of X. What mass of Y will it contain?

(1) 60 g (2) 80 g

(3) 30 g (4) 24 g

17. Match the compounds given in Column I with % composition of oxygen given in Column II.

Use: Atomic mass (in u): C = 12, O = 16, S = 32, Cl = 35.5, Cr = 52.

Column-I		Column-II	
A	CrO_5	P	47.1%
B	SO_3	Q	18.4%
C	C_3O_2	R	60.6%
D	Cl_2O	S	60.0%

	A	B	C	D
(1)	P	Q	S	R
(2)	R	S	Q	P
(3)	P	S	Q	R
(4)	R	S	P	Q

18. Mass remains conserved in a chemical reaction is a statement of:

(1) Law of combining volumes of gases

(2) Law of conservation of mass

(3) Law of multiple proportion

(4) Law of definite proportion

19. Element Q has 20.0% of ^{62}Q and 80.0% of ^{64}Q. The average atomic mass (amu) is closest to:

(1) 62.4 (2) 63.2

(3) 63.6 (4) 64.0

20. 6 gram molecule of ozone contains the number of electrons equal to that in

(1) 0.5 mol of hydrogen atom

(2) 9 mol of sulphur atom

(3) 7 mol of nitrogen atom

(4) 2.3 mol of sodium atom

teger Type Questions

A 0.20 mole sample of a hydrocarbon C_xH_y after complete combustion with excess of O_2 gas yields 0.80 mole of CO_2 and 1 mole of H_2O. Hence, x + y is

2. What volume (in litre) of hydrogen (at STP) will be liberated on reaction of Zn (excess) with 50 ml dilute H_2SO_4 of specific gravity 1.3 and having 40% H_2SO_4 by mass? (Round off to nearest integer) ($Zn + H_2SO_4 \rightarrow Zn\,SO_4 + H_2$)

23. Two elements A and B form 0.15 moles of each A and AB_3 type compounds. If both A_2B and A weigh equally, then the atomic weight of A is times of atomic weight of B.

24. If the molarity of H^+ ion obtained after mixing of 20 ml of 0.05 M H_3PO_4 and 100 ml of 0.1 M of Ca(OH is $\frac{1}{x}$ M. Find the value of x.

25. The volume of 10 M HCl solution required to prepa a 200 mL solution of 0.4 M HCl is y m.c. Find y.

ANSWER KEY

1.	(1)	6.	(2)	11.	(2)	16.	(1)	21.	(14)
2.	(1)	7.	(1)	12.	(2)	17.	(4)	22.	(6)
3.	(2)	8.	(3)	13.	(2)	18.	(2)	23.	(2)
4.	(1)	9.	(2)	14.	(1)	19.	(3)	24.	(30)
5.	(4)	10.	(2)	15.	(1)	20.	(2)	25.	(8)

HINT & SOLUTIONS

1. **(1)**
% free SO_3 in 109% oleum = 40%% free SO_3 in 118 % oleum = 80%

SO_3	+	H_2O	→	H_2SO_4
40g		9g		
80g		18g		

$\Rightarrow$ ratio of % of free SO_3 = 40 : 80 = 1 : 2

2. **(1)**
First, calculate the molar masses:

$M_{VO} = 51+16 = 67$ g/mol,

$M_{Fe_2O_3} = 2(56)+3(16) = 160$ g/mol, and

$M_{FeO} = 56+16 = 72$ g/mol.

Next, find the number of moles of each reactant:

$$\text{Moles of VO} = \frac{6.7\text{ g}}{67\text{ g/mol}} = 0.1\text{ mol.}$$

$$\text{Moles of } Fe_2O_3 = \frac{4.8\text{ g}}{160\text{ g/mol}} = 0.03\text{ mol.}$$

The reaction is $2VO + Fe_2O_3 \rightarrow 2FeO + V_2O_5$. The stoichiometric ratio requires 2 moles of VO for every 1 mole of Fe_2O_3.

To react with 0.1 mol of VO, we would need $0.1/2 = 0.05$ mol of Fe_2O_3. Since we only have 0.03 mol, Fe_2O_3 is the limiting reactant.

According to the stoichiometry, 1 mole of Fe_2O_3 produces 2 moles of FeO.

Moles of FeO produced =

$$0.03\text{ mol } Fe_2O_3 \times \frac{2\text{ mol FeO}}{1\text{ mol } Fe_2O_3} = 0.06\text{ mol.}$$

Weight of FeO produced = $0.06\text{ mol} \times 72\text{ g/mol} = 4.32\text{ g.}$

3. **(2)**
Let's derive the relationship by considering 1 L of the solution.

$$\text{Molarity (M)} = \frac{\text{moles of solute}}{\text{Volume of solution (L)}}.$$

So, in 1 L of solution, moles of solute = M.

Molar mass of solute is M_2.

Mass of solute in 1 L of solution = $M \times M_2$ (in grams).

Density of solution is $d(\text{in g/mL})$. Mass of 1 L (1000 mL) of solution = Volume × Density = $1000 \times d$ (in grams).

Mass of solvent = Mass of solution - Mass of solute = $(1000d - M \cdot M_2)$ g.

Mass of solvent in kg = $\frac{1000d - M \cdot M_2}{1000}$ kg.

Molality (m) = $\frac{\text{moles of solute}}{\text{Mass of solvent (kg)}}$

$= \frac{M}{\frac{1000d - M \cdot M_2}{1000}} = \frac{1000 \cdot M}{1000d - M \cdot M_2}$.

This expression matches option (2), which is $m = \frac{M}{1000d - MM_2} \times 1000$.

(1)

$N_2(g) + 3H_2 \rightarrow 2NH_3$

1 mole N_2 required 3 moles of H_2.

(a) N_2 : H_2

0.5 mol : 2 mol

N_2 is L.R.

(b) $\frac{28}{28}$: $\frac{6}{2}$

1 mol : 3 mol

No L.R

(c) $\frac{56}{28}$: $\frac{10}{2}$

2 mol : 5 mol

So, H_2 is L.R.

(d) N_2 : H_2

84 mg : 18 mg

$= 3 \times 10^{-3}$ mol : 9×10^{-3} mol

No L.R.

(4)

No. of molecules = $\frac{1.2 \times 2 \times N_A}{35 \times 70}$

(2)

First, calculate the moles of urea. Using Avogadro's number ($N_A \approx 6.02 \times 10^{23}$ mol^{-1}).

Moles = Number of molecules / $N_A = (6.02 \times 10^{20}) / (6.02 \times 10^{23}) = 10^{-3}$ mol.

Next, convert the volume of the solution to liters: Volume = 100 mL = 0.1 L.

Finally, calculate the molarity (M), which is moles of solute per liter of solution.

Molarity = Moles / Volume (L) = 10^{-3} mol / 0.1 L = 10^{-2} M = 0.01 M.

7. **(1)**

The balanced equation is $2A + 3B \rightarrow X + 2Y$. We start with 2 moles of A and 1 mole of B. The stoichiometric ratio requires 2 moles of A for every 3 moles of B.

To react with 1 mole of B, we need $(1 \text{ mol B}) \times (2 \text{ mol A} / 3 \text{ mol B}) = 2/3$ mol of A.

Since we have 2 moles of A (which is > 2/3 mol), B is the limiting reactant.

The reaction consumes all 1 mole of B.

Moles of A reacted = $1 \times (2/3) = 2/3$ mol.

Moles of A left = $2 - 2/3 = 4/3$ mol. (Matches S)

Moles of B left = $1 - 1 = 0$ mol. (Matches R)

Moles of X formed = $1 \times (1/3) = 1/3$ mol. (Matches P)

Moles of Y formed = $1 \times (2/3) = 2/3$ mol. (Matches Q)

Therefore, the correct match is I-S, II-R, III-P, IV-Q.

8. **(3)**

First calculate the percentage of C and H as:

% of C

$= \frac{\text{Wt of } CO_2}{\text{Wt of comp.}} \times \frac{12}{44} \times 100$

$= \frac{5.28}{3.60} \times \frac{12}{44} \times 100 = 40\%$

% of H

$= \frac{\text{Wt of } H_2O}{\text{Wt of comp.}} \times \frac{2}{18} \times 100$

$= \frac{2.16}{3.60} \times \frac{2}{18} \times 100 = 6.67\%$

Now, % of O = 100 – (40 + 6.67) = 53.33%

9. **(2)**

$2y + y + 1.34 + 42.65 = 100$

$y = 18.67$

$x = 37.34$

Element	C	N	H	O
%	37.34	18.67	1.34	42.65
Moles	$\frac{37.34}{12}$	$\frac{18.67}{14}$	$\frac{1.34}{1}$	$\frac{42.65}{16}$
	= 3.111	1.333	1.34	2.665
Simple ratio	= 2.33	1	1	2] × 3
	= 7	3	3	6

EF = $C_7H_3N_3O_6$;

$n = \frac{450.22}{225} = 2$; MF = $C_{14}H_6N_6O_{12}$

10. **(2)**

Let the moles of $FeSO_4$ be n_1 and the moles of $Fe_2(SO_4)_3$ be n_2.

Moles of SO_4^{2-} from $FeSO_4 = n_1$.

Moles of SO_4^{2-} from $Fe_2(SO_4)_3 = 3n_2$.

Given that the number of sulphate ions is equal:

$n_1 = 3n_2$.

Moles of Fe^{2+} from $FeSO_4 = n_1$.

Moles of Fe^{3+} from $Fe_2(SO_4)_3 = 2n_2$.

The required molar ratio is $Fe^{2+} : Fe^{3+} = n_1 : 2n_2$.

Substitute $n_1 = 3n_2$ into the ratio:

Ratio $= 3n_2 : 2n_2 = 3:2$.

11. **(2)**

$2KClO_3 \rightarrow 2KCl + 3O_2$

48 g of O_2 would be produced from pure $KClO_3$

$= \frac{2\times122.5}{96}\times 48$

80% $KClO_3$ required $= \frac{2\times122.5}{96}\times 48\times\frac{100}{80}$

$= 153.12$ g

12. **(2)**

Theoretically 244g Sb give 340 g Sb_2S_3

$= 2$ g of Sb should give $\frac{340}{244}\times 2 = 2.78$g of Sb_2S_3

% yield $= \frac{1.100}{2.78} \simeq 36\%$

13. **(2)**

Let x = moles of ZnS. From yield:

Zn produced $= x \times 0.8 \times 0.8 = 0.64x$

Set $0.64x = 3.2 \Rightarrow x = 5.0$

14. **(1)**

$m = \frac{\chi_B}{1-\chi_B}\times\frac{1000}{M_A} = \frac{0.2}{1-0.2}\times\frac{1000}{18} = 13.88$ m

m = molality, χ_B = mole fraction of solute.

15. **(1)**

1 amu equals 1.66×10^{-24} g.

Therefore, the mass of one calcium atom is

$40 \times 1.66 \times 10^{-24} = 6.64 \times 10^{-23}$ g.

16. **(1)**

(16/24) = (40/Y); Y = 60 g.

17. **(4)**

Percentage of Oxygen in $CrO_5 = 80/132 = 60.6\%$

Percentage of Oxygen in $SO_3 = 48/80 = 60\%$

Percentage of Oxygen in $C_3O_2 = 32/68 = 47.1\%$

Percentage of Oxygen in $Cl_2O = 16/87 = 18.4\%$

18. **(2)**

Law of definite proportion

Example: $CuCO_3$ natural and synthetic sample has same percentage of Cu,C,O.

19. **(3)**

$M_{avg} = 0.20 \times 62 + 0.80 \times 64 = 12.4 + 51.2 = 63.6$ amu.

20. **(2)**

6 grams molecular $O_3 = 6$ mol O_3

No. of e^- in 6 mol $O_3 = 6 \times (8\times3)\, N_A = 144\, N_A$

9 mol of $S = 9 \times 16\, N_A e^- = 144\, N_A e^-$

21. **(14)**

$$\underset{0.20\text{ mole}}{C_xH_y} + \left(x+\frac{y}{4}\right)O_2 \longrightarrow \underset{0.80}{xCO_2} + \frac{y}{2}H_2O$$

$20x = 0.80 \Rightarrow x = 4$ 1 mole

$0.2y = 2 \Rightarrow y = 10$ $\frac{y}{2} = 1$ mole

$y = 2$ mole

Hence Hydrocarbon is C_4H_{10}

22. **(6)**

Mass of $H_2SO_4 = \frac{50\times1.3\times40}{100}$

$= 26$ g

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

Volume of H_2 at STP $= \frac{22.4}{98}\times 26$

≈ 6 L

23. **(2)**

Let the mass of A_2B and AB_3 is 1gm and atomic mass A and B are x and y respectively.

$\therefore \quad 0.15 = \frac{1}{2x+y}$, and $0.15 = \frac{1}{x+3y}$,

On solving we get

$x = 2y$

24. **(30)**

No. of equivalents of $H_3PO_4 = 0.2 \times 0.05 \times 3 = 0.03$

No. of equivalents of $Ca(OH)_2 = 0.1 \times 0.1 \times 2 = 0.02$

No. of equivalents of H_3PO_4 left $= 0.03 - 0.02 = 0.01$

Normality $= \frac{0.01}{300}\times 1000 = \frac{1}{30}$

$x = 30$

25. **(8)**

$M_1V_1 = M_2V_2$

$10 \times V_1 = 0.4 \times 200$

$V_1 = 8$ mL